Networking

- **Networking Essential**
  OSI Model (open system interconnection)
  Developed by ISO that describe how communication should occur in a computer network.
  OSI model composed of:
  1. Physical layer (physical connection between devices) e.gcables
  2. Data Link Layer (Medium to transmit signal) e.g. Ethernet
  3. Network Layer(sending data between different network) e.g IP
  4. Transport Layer (End to end communication on different host) Tcp
  5. Session Layer(Initiating communication between applications)NFS
  6. Presentation Layer(Data is delivered in a form application understands)
  7. Application layer(End user application)

Layer Number	Layer Name	Main Function	Example Protocols and Standards
Layer 7	Application layer	Providing services and interfaces to applications	HTTP, FTP, DNS, POP3, SMTP, IMAP
Layer 6	Presentation layer	Data encoding, encryption, and compression	JPEG, PNG, MPEG
Layer 5	Session layer	Establishing, maintaining, and synchronising sessions	NFS, RPC
Layer 4	Transport layer	End-to-end communication and data segmentation	UDP, TCP
Layer 3	Network layer	Logical addressing and routing between networks	IP, ICMP, IPSec
Layer 2	Data link layer	Reliable data transfer between adjacent nodes	Ethernet (802.3), WiFi (802.11)
Layer 1	Physical layer	Physical data transmission media	Electrical, optical, and wireless signals

TCP/IP MODEL (Transmission Control protocol/Internet Protocol)

It comprises of:

- Application(Application layer)
- Transport-(Transport layer)
- Network-(Internet layer)
- Link-(

IP Addresses and subnets

Every host on the internet needs a unique identifier for the host to communicate with .

Ip address comprises four octets i.e 32 bits an octet allows us to represent a decimal number between 0-255.

0 and 255 are reserved for the network and broadcast addresses e.g 192.168.1.0 is network address,192.168.1.255 is broadcast address. Broadcast address targets all the hosts on the network.

Commands used to configure networks.

Ifconfig(shows ip address

```
Terminal
user@TryHackMe$ ifconfig
[...]
wlo1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.66.89  netmask 255.255.255.0  broadcast 192.168.66.255
        inet6 fe80::73e1:ca5e:3f93:b1b3  prefixlen 64  scopeid 0x20<link>
        ether cc:5e:f8:02:21:a7  txqueuelen 1000  (Ethernet)
        RX packets 19684680  bytes 18865072842 (17.5 GiB)
        RX errors 0  dropped 364  overruns 0  frame 0
        TX packets 14439678  bytes 8773200951 (8.1 GiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Types of ip addresses

Private Addresses-(used only inside local networks.)192.168.1.5

Public address(-an address that is reachable over the internet.)8.8.8.8

IP PROTOCALS

UDP-(user Datagram Protocol)allows us to reach specific process on the target host

Its connectionless protocol- it doesn't need established connection

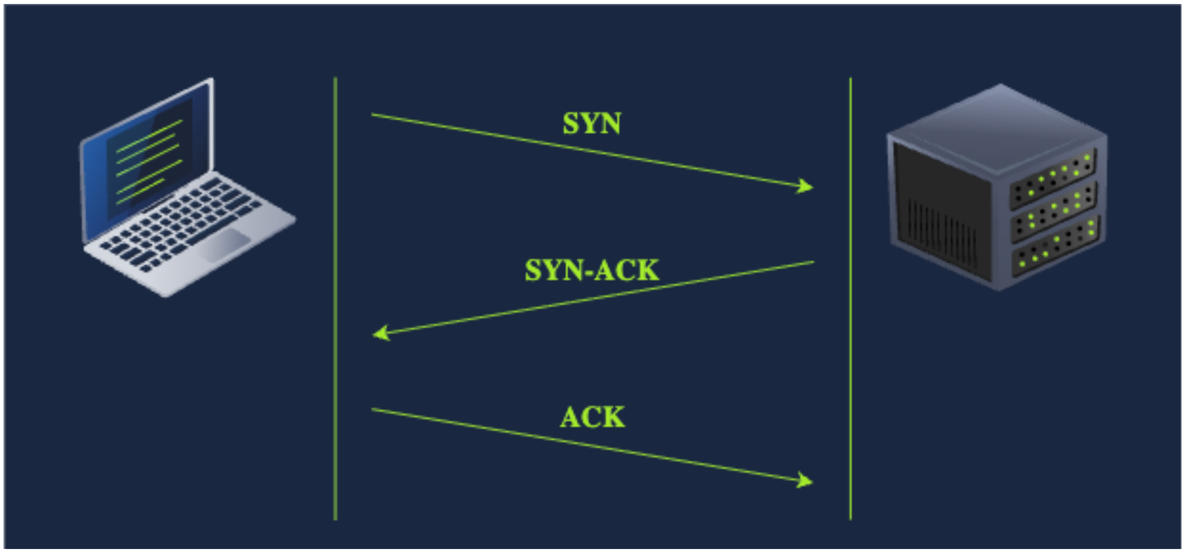
Example sms(standard mail service)no delivery confirmation

TCP-**(Transmission control protocol)-**connection oriented protocol, it requires establishment of tcp connection before data is sent.3 way handshake

SYN Pcket:client initiate connection

SYN-ACK packet-server respond to SYN

ACK packet-handshake is completed



ENCAPSULATION

It’s a process of every layer adding a header to the received unit of data .

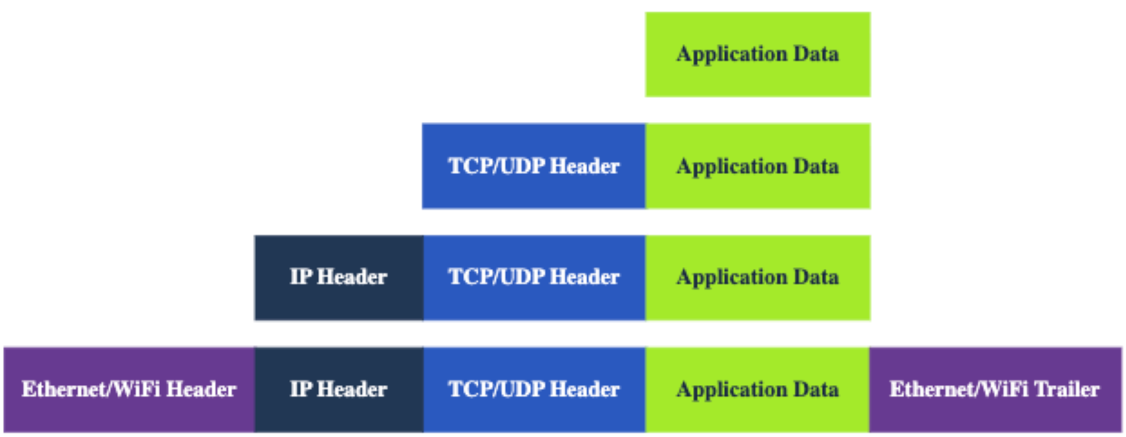
Steps:

Application data-user input the data they want to send into the application

Transport-TCP or UDP add proper header

Network packet-internet layer add ip header to received tcp segment or udp

Data link-ethernet receives ip packet and add header trail creating frame.



TELNET(Teletype network)

Network protocol for remote terminal connection

protocol	Transport Protocol	Default Port Number
TELNET	TCP	23
DNS	UDP or TCP	53
HTTP	TCP	80
HTTPS	TCP	443
FTP	TCP	21
SMTP	TCP	25
POP3	TCP	110
IMAP	TCP	143